INCH-POUND

MS27511L
w/AMENDMENT 1
27 June 2025
SUPERSEDING
MS27511L
2 February 2010

DETAIL SPECIFICATION SHEET

COVERS, PROTECTIVE, ELECTRICAL, CONNECTOR RECEPTACLE, BAYONET COUPLING, SERIES II

Reactivated after 10 June 2005 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

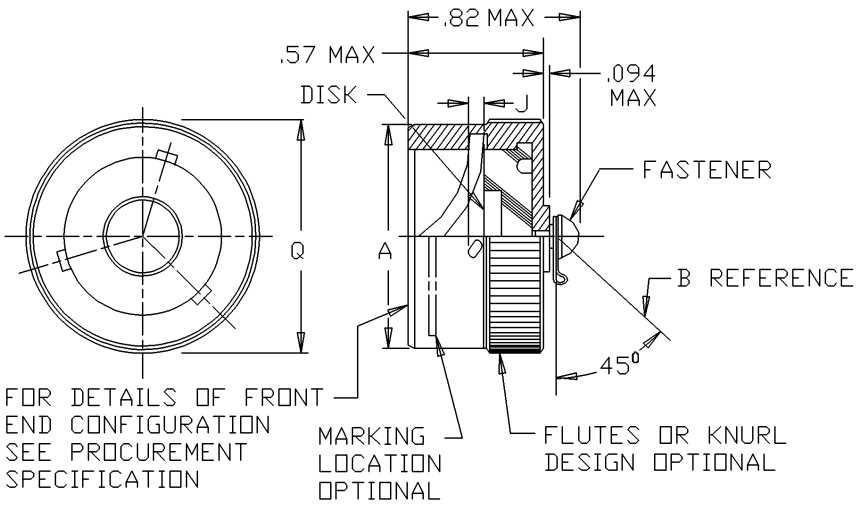


FIGURE 1. Cover.

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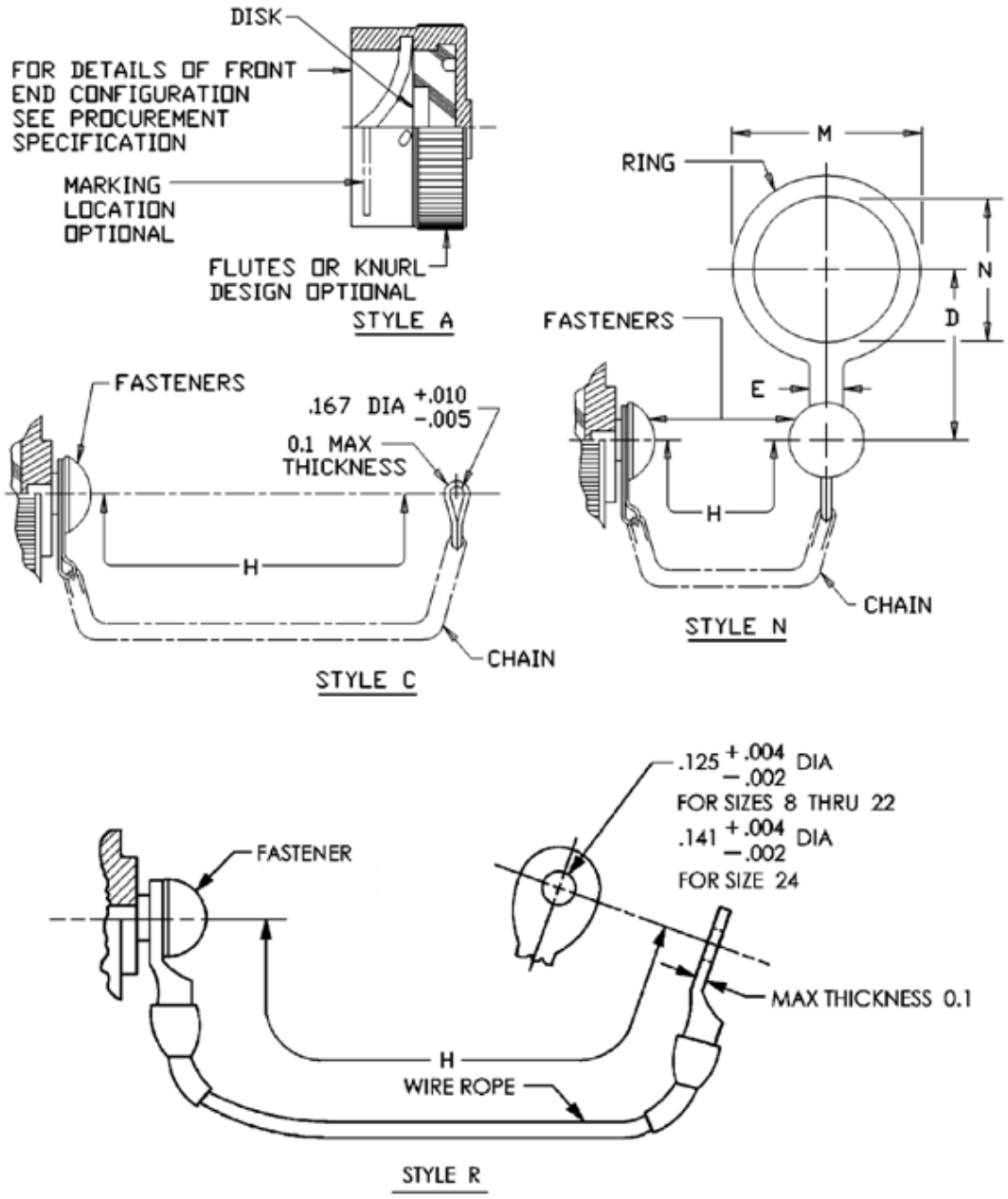


FIGURE 1. Cover – Continued.

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Shell size	A dia max	D		E min ring throat	H +.500 -.250 standard chain or rope length	J ±.015	M max dia	N min dia	Q max dia
		max	min						
8	.630	1.13	.80	.23	3.0	.082	1.22	.89	.750
10	.761	1.23	.93				1.42	1.01	.859
12	.936	1.29	.99		3.50		1.55	1.14	1.031
14	1.074	1.35	1.05				1.68	1.26	1.156
16	1.172	1.41	1.11				1.80	1.39	1.281
18	1.304	1.47	1.18				1.92	1.51	1.406
20	1.435	1.54	1.24		4.0		2.05	1.64	1.531
22	1.560	1.60	1.30				2.18	1.76	1.641
24	1.688	1.66	1.36			.107	2.30	1.89	1.766

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	.05	.57	14.5	1.11	28.2	1.36	34.5	1.66	42.2
.004	.10	.630	16.00	1.13	28.7	1.39	35.3	1.68	42.7
.005	.13	.750	19.05	1.14	29.0	1.406	35.71	1.688	42.87
.010	.25	.761	19.33	1.156	29.36	1.41	35.8	1.76	44.7
.015	.38	.80	20.3	1.172	29.77	1.42	36.1	1.766	44.86
.030	.76	.82	20.8	1.18	30.0	1.435	36.45	1.80	45.7
.094	2.39	.859	21.82	1.22	31.0	1.47	37.3	1.89	48.0
.107	2.72	.89	22.6	1.23	31.2	1.51	38.4	1.92	48.8
.1	3.0	.93	23.6	1.24	31.5	1.531	38.90	2.05	52.1
.125	3.18	.936	23.77	1.26	32.0	1.54	39.1	2.18	55.4
.141	3.58	.99	25.1	1.281	32.54	1.55	39.4	2.30	58.4
.167	4.24	1.01	25.7	1.29	32.8	1.560	39.62	3.00	76.2
.23	5.8	1.031	26.19	1.30	33.0	1.60	40.6	3.50	88.9
.25	6.4	1.05	26.7	1.304	33.12	1.64	41.65	4.00	101.6
.50	12.7	1.074	27.28	1.35	34.3	1.641	41.68		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Finish: See MIL-DTL-38999.
4. Not for space born applications.

FIGURE 1. Cover – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Interface dimensions shall conform to MIL-DTL-38999.

Material:
Ring - Passivated stainless steel, .030 maximum thickness.
Cover - Impact extruded or machined aluminum alloy.
Disk - Silicone or fluorosilicone elastomer.
Chain - Passivated stainless steel, conforming to RR-C-271, type II, class 3.
Wire rope - Insulated, braided steel wire.
Fastener - Passivated stainless steel.

Eyelets shall be crimped over wire rope as to ensure a breaking strength equal to or greater than that of the wire rope.
Mating connectors: MIL-DTL-38999 Series II receptacles and MIL-DTL-27599 Series II receptacles.
Disk shall be bonded to cover, or mechanically retained.
The chain or rope assembly shall rotate freely on the cover fastener.
The protective cover and rope assembly shall withstand a static tensile load of 35 pounds applied along the B reference line for 5 minutes.
Fastener tensile strength: Both axial and longitudinal directions shall be 25 pounds. The load shall be applied to the end of the chain or wire rope.

Qualification:

Examination of product.
Mating and unmating forces.
Humidity.
Salt spray (corrosion).
Fastener tensile strength (see requirement).
Altitude immersion.

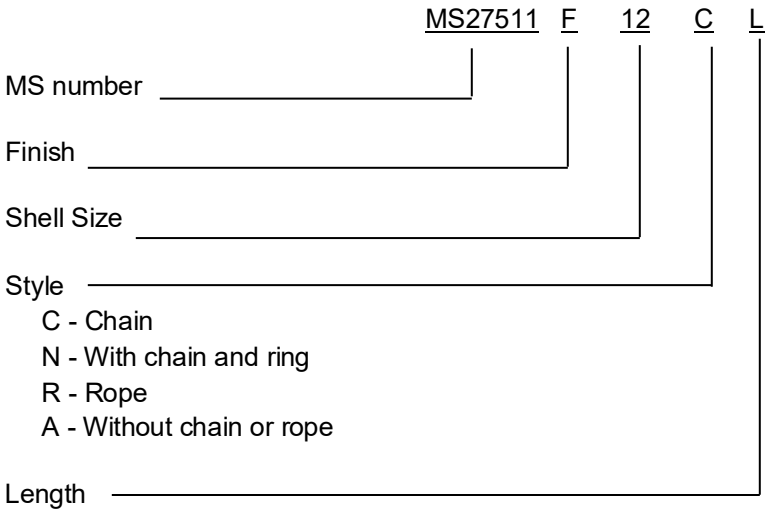
Group A sampling inspection. 100% inspection is not applicable. Sampling inspection shall be as specified in table I:

TABLE I. Group A sampling plan.

Lot size	Sample size
1 to 13	100 percent
14 to 150	13 units
151 to 280	20 units
281 to 500	29 units
501 to 1,200	34 units
1,201 to 3,200	42 units

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Part or Identifying Number (PIN) example:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-27599
RR-C-271

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2025-050)

Review activities:
Army – AR, MI
Navy – EC, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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